

Integrating Landscape Accumulation Functions and Sewer Networks to Identify Salt Vulnerable Areas in a Highly Urbanized Watershed

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Road Salt & Freshwater Salinization

Rapid urbanization in cold climates has driven an increase in the application of road salt (NaCl) to maintain safe winter driving conditions.

Road salt is known to be a significant driver of freshwater salinization (Dugan et al., 2017; Kaushal et al., 2023).

- Chloride from road salt is mobilized during rainfall and snowmelt events, raising the chloride concentration in streams (Kaushal et al., 2023).
- Salinization can have impacts on aquatic organisms, drinking water, and infrastructure (Kaushal et al., 2021).

Kaushal et al., (2021), identified a significant need for input-based predictive models alongside frequent monitoring of salt and chemical cocktails in freshwater to manage the continuing impacts of freshwater salinization syndrome (FSS) as salt inputs increase.

Salt Vulnerable Areas (SVAs) are ; "... an area particularly sensitive to road salts where additional salt management measures may be necessary to mitigate the environmental effects of road salts in that area" (ECCC, 2004).

- This project aims to develop a robust system for predicting chloride hazard in urban freshwater systems based on impervious surface data and direct flow modelling.*

Study Area

- The Black Creek is a tributary of the Humber river in the west end of Toronto, Ontario & parts of York Region.
 - The Black Creek Watershed is highly urbanized catchment area.
 - 41.7%** pervious surfaces.
 - 40.6%** impervious surfaces including roads, parking.
 - 17.7%** buildings.

The extent of urbanization in the Black Creek watershed as well as the varying age and capacity of its infrastructure makes it an ideal candidate for studying urban hydrology.

- Black Creek has an extensive sewer network including sanitary and stormwater infrastructure.
 - The southern end of the watershed contains a significant portion of combined sewer networks.
 - ~200 outfalls drain storm and combined water into the stream.

- Sewer networks can be observed extending outside of the watershed boundary (Figure 1).
 - These sewer lines are connected to Outfalls inside the watershed, essentially extending the range of drainage.

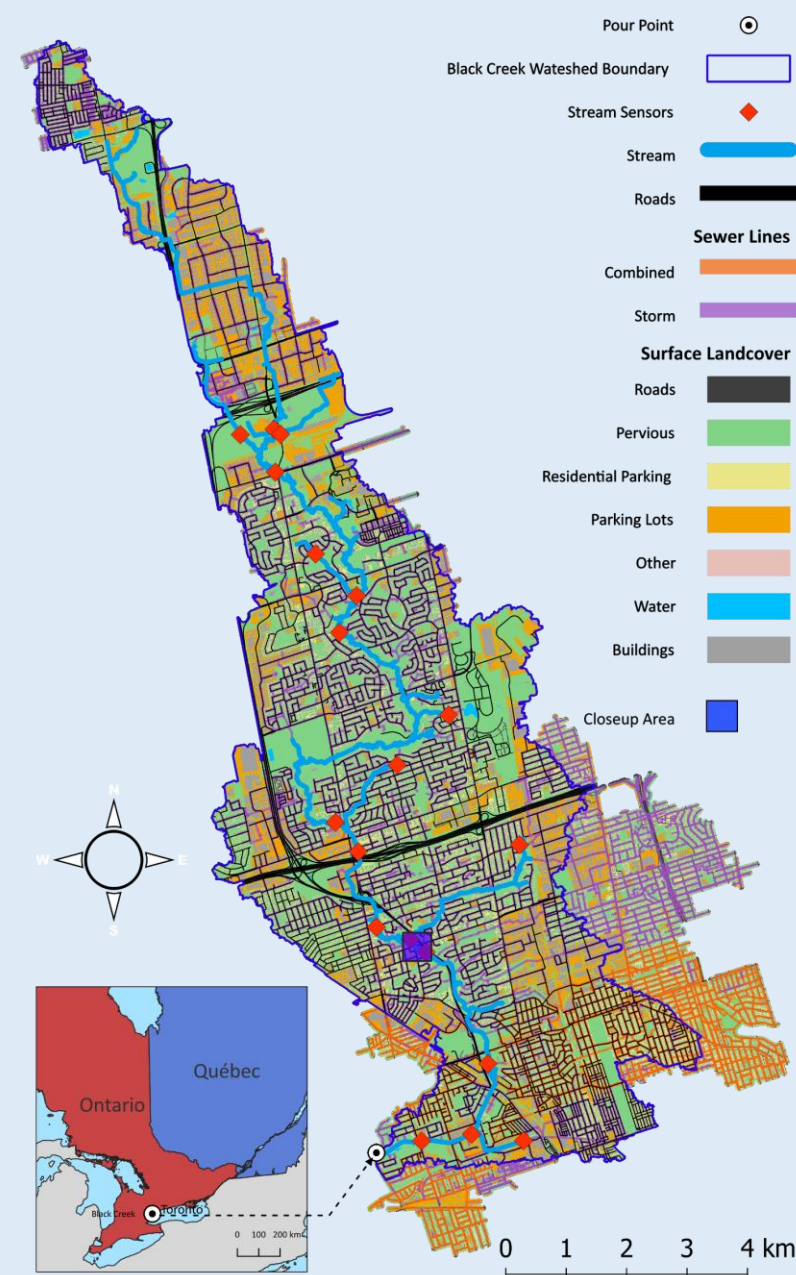


Figure 1. Black Creek watershed and its location in Southern Ontario, with roads and stormwater sewer lines, land cover, and locations of in-stream Levellogger LTC sensors.

Methodology & Flow Connectivity

Landscape Accumulation Functions (LAFs) (Richardson and Foster 2024) are a flow-path-aware method of determining the influence of nearby landcover on rivers and streams.

- LAFs take a Digital Elevation Model (DEM) and categorical landcover rasters as inputs and calculate flow accumulation over the given categorical variables (Figure 2).
- The landcover accumulation (calculated using a mass-flux formula) is compared with overall flow accumulation to determine the proportion of upstream cover of each categorical variable for every cell in the raster.
- LAFs are implemented using the open-source 'thalweg' python library (Foster, 2025).

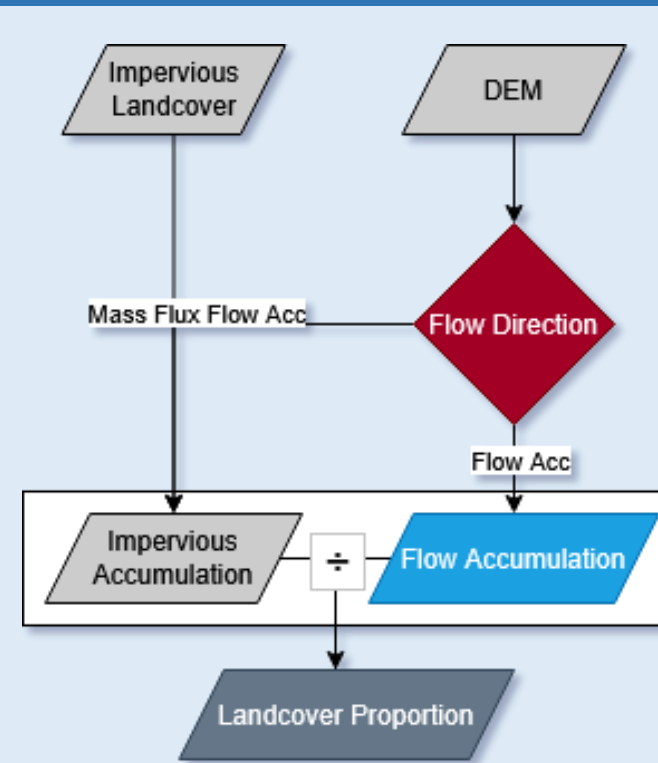


Figure 2. A representation of the overall process of LAFs.

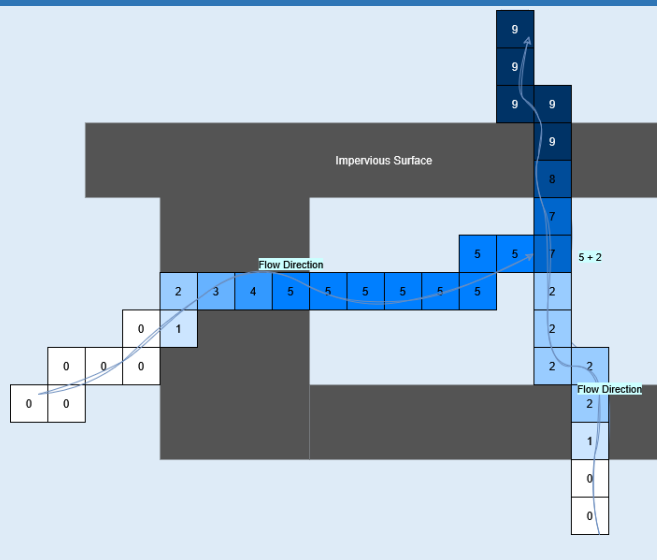


Figure 3. The LAF uses a mass-flux flow accumulation algorithm to approximate the influence the impervious surfaces on the overall flow to the stream. The categorical 'Impervious Surface' raster is considered the 'loading' layer, whereby accumulation only occurs over an impervious surface. The values in each cell, when divided by the overall flow accumulation, yield the upstream landcover proportion.

Twenty Solinst Levellogger LTC sensors placed in-stream throughout the Black Creek at key locations record electrical conductivity at 15-minute intervals.

- Electrical Conductivity (EC) ($\mu\text{S}/\text{cm}$) is used as a surrogate for chloride concentration in the stream.

The landcover proportion (LCP) is sampled at the location of each LTC sensor in the stream, then compared with the relevant EC values.

- The Pearson Correlation Coefficient is calculated for each LCP-EC pair at various timescales.
- Conductivity samples are averaged by week and overall.

A high r-value indicates that the upstream impervious surface proportion has an influence on the in-stream chloride concentrations.



Figure 4. A closer look at a section of the Black Creek, representing sewer infrastructure, landcover classes, and sensors.

Impervious Surfaces

Land use has an impact on neighbouring aquatic ecosystems, and there is a relationship between the distance of a land use type from an aquatic system and its effect on the system (Weller et al., 2023).

Existing methods of quantifying this influence such as Landscape Configuration Metrics (LCM) and Spatial Stream Networks (SSN) do not account for complexities in flow.

In this project, surface landcover is represented in a 1m x 1m resolution land-use classified raster (Figure 3).

- Impervious surfaces, such as Roads, Large Parking Lots, and Residential Driveways are salted during winter months, and are known to be a major contributor to chloride concentrations in streams during flow events (Oswald et al., 2023).

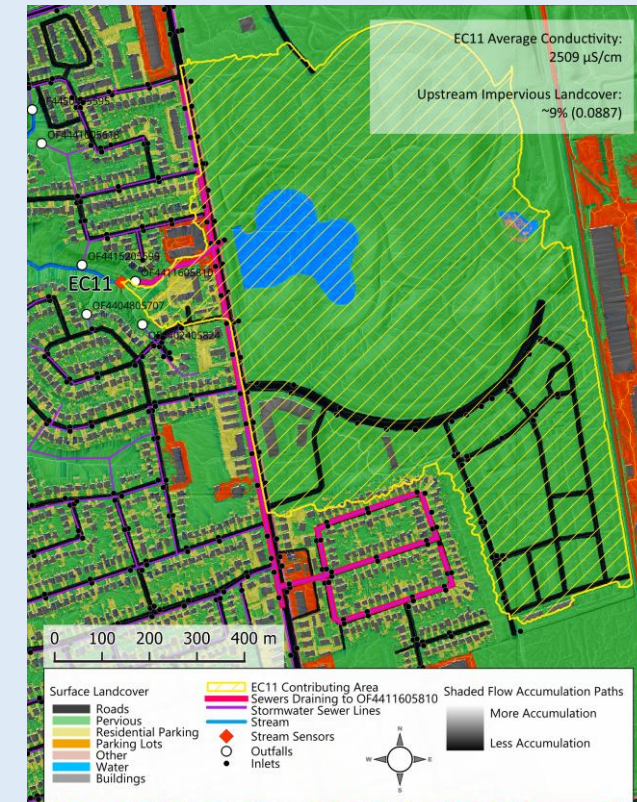


Figure 5. Flow paths are illustrated by calculating the log of the overall accumulation values. Impervious surfaces accumulate only on raster cells that overlap with the Impervious Surfaces layer, which is overlaid in blue.

The value at each cell for the accumulated raster is the number of upstream cells that contribute to it.

The impervious landcover proportion (LCP) can also be described as the percent of cells upstream of each location which are covered by an impervious surface.

The proportion is calculated by dividing the impervious landcover accumulation raster by the overall flow accumulation raster.



- Recently-installed sewer infrastructure and footpaths are not present in the data at this point, leading to inaccuracies in the flow accumulation model.
- EC11 is considered an outlier at this point and is therefore removed from the correlation analysis.

The correlation between LCP and EC for the average conductivity is moderate to high when sensor EC11 is excluded from the analysis (Figure 7, Figure 8).

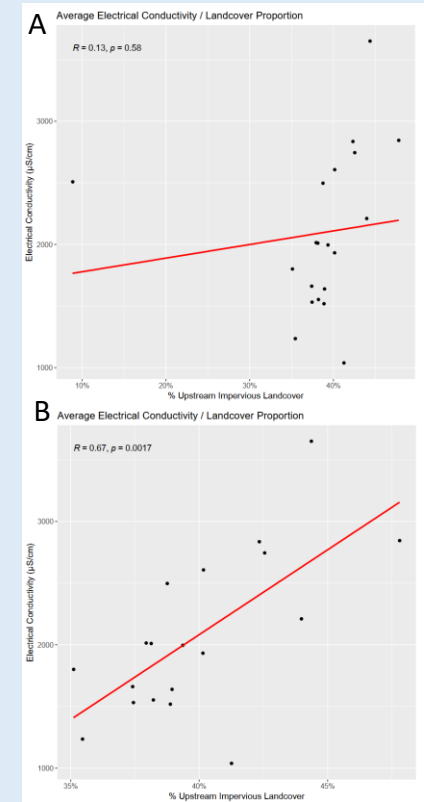


Figure 8. Both charts show the correlation between landcover proportion (LCP) - x-axis, and Electrical Conductivity (EC) - y-axis, as well as the Pearson Correlation Coefficient for the average conductivity across all timestamps/samples. (A) Every sensor: R = 0.13, p = 0.58. A significant outlier is noted on the left of the x-axis. (B) Excluding sensor EC11: R = 0.67, p = 0.0017.

Analysis

The Pearson Correlation Coefficient between EC and LCP for each sensor can also be calculated at each timestamp or averaged over any given period.

- Only timestamps where samples for at least **10** sensors were recorded were included.

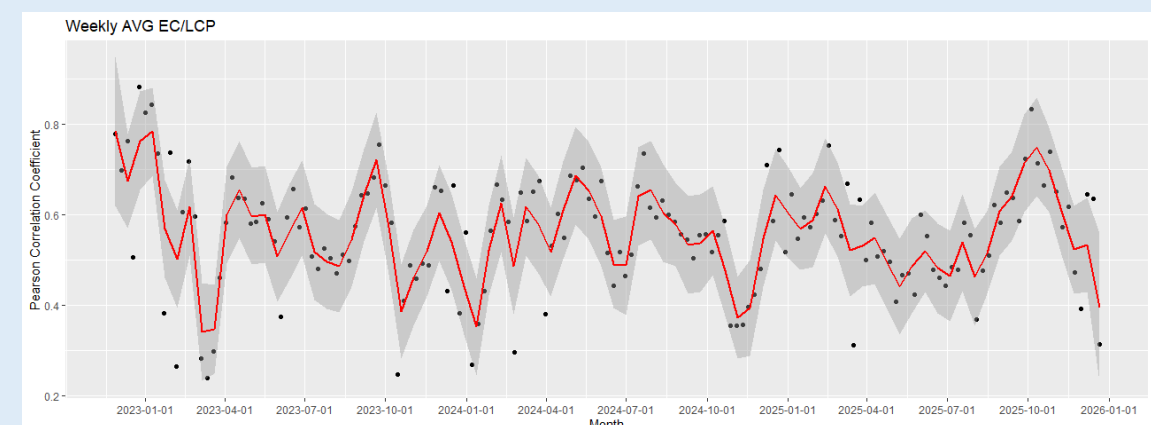


Figure 10. The correlation between LCP and EC over time for conductivity averaged by week. (Charts for daily and rolling averages were also made but were too busy to easily interpret).

Maximum correlation (AVG Daily): R = 0.897 | February 17th 2023

Minimum correlation (AVG Daily): R = -0.238 | January 18th 2023

Median correlation (AVG Daily): R = 0.543 | October 5th 2023

(Figure 10) shows significant variation of the correlation coefficient over time, with distinct peaks and troughs at different times of the year and clear month-to-month changes. Clear peaks and drops can be seen between January - April 2023, and August - January 2026.

- A more in-depth analysis of weather and snowmelt events may be required to fully explain the variation.

Continuing Work

Subsurface flow pathways are frequently left out of watershed-scale studies, which can heavily misrepresent real flow in areas with extensive networks (Oswald et al., 2023).

- The LAF model presented at this stage of the study does not include stormwater drainage systems.
- Stormwater drainage will be considered at a future stage of the project through the implementation of and Adaptive Stormwater Infrastructure (ASI) flow accumulation algorithm (Choi et al., 2011).

The inconsistencies noted in the example of sensor EC11 (Figure 7) are a clear indicator of the need to include stormwater infrastructure in flow modelling of heavily urbanized areas.

- The validity of any model which implements such data will depend on the quality of the input data.
- An open-source python library is being developed alongside the project to manage sewer datasets, derive networks, and implement the ASI algorithm.

A significant amount of future work will involve understanding the correlation patterns observed in (Figure 10), and better determining the relationships between impervious surfaces, chloride accumulation risk, and Salt Vulnerable Areas.

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